

Project 2 (14b)

Tasks

- Image Classification - 10-classes min
- Segmentation - multiclass/semantic segmentation
- Localization - X/Y (1 object, center)
- Pose estimation - Keypoint localization
- Encoder-Decoder
- Image synthesis

Project 2 (9b)

- Requirements

- Data analysis
- Data preprocessing and normalization
- Data split
- Augmentations
- Configuration
- Experiment tracking
- Experiments - meaningful based on the results of previous experiments
 - Including improvement techniques (e.g. Dropout, Normalization layers, Skip Connections, Bottleneck Layers, ...)
- Results and evaluation metrics
- Clear code
- Markdown documentation and comments
- Final presentation of projects
- Effort on consultations

Project 2

- Consultation 1 - minimal requirements
 - Select task ([google sheet](#) maximum 2 teams per task, first come, first serve)
 - Download and examine dataset
 - Exploratory data analysis ([example](#))
 - Select evaluation metrics
 - Preprocessing and initial augmentations

Project 2 - List of topics

Animal 10 - classification

- <https://www.kaggle.com/datasets/alessiocorrado99/animals10>

Sport classification

- <https://www.kaggle.com/datasets/gpiosenka/sports-classification>

LabelMe-12 - classification

- <https://www.ais.uni-bonn.de/download/datasets.html>

Traffic Lights - localization

- <https://www.kaggle.com/datasets/mbornoe/lisa-traffic-light-dataset>

Lizard - segmentation

- <https://www.kaggle.com/datasets/aadimator/lizard-dataset>

Tiger - segmentation

- <https://tiger.grand-challenge.org/>

American Sign Language - classification

- <https://www.kaggle.com/datasets/kapillondhe/american-sign-language>

CelebA - image synthesis

- <https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>

Facial keypoint localization (Regression or Heatmaps)

- <https://www.kaggle.com/c/facial-keypoints-detection/data>

Custom **dataset/problem** allowed after:

- Submit dataset/problem info within 1 week of project assignment.
- Problem must be image-based.
- Approval by instructor required.